

Blood Donation Management System - Software Requirement Specification (SRS)

1. Project Overview

Objective: Develop a mobile application and web admin portal that connects blood donors and receivers.

2. User Roles

- Donor
- Receiver
- Admin

3. Authentication Module

- Mobile OTP Login
- Email Verification
- Aadhaar Verification

4. Registration Module

Fields:

Full Name, Mobile Number, Email, Aadhaar Number, Blood Group, City, Area

5. Role Selection

Roles:

- Donor
- Receiver

Role change requires Aadhaar + OTP verification.

6. Donor Profile Setup

Fields:

DOB, Gender, Last Donation Date, Availability

7. Donor Dashboard

Displays:

Total Donations, Lives Saved, Nearby Requests

8. Nearby Requests Module

Filters:

Blood Group, Distance, City

Urgency:

Medium, High, Urgent, Critical

9. Request Details

Patient Info, Hospital Info, Contact Info, Location, Call Now, I Can Donate

10. Donation Confirmation Flow

Request Acceptance and Notification Flow

11. Donation History

All / Donated / Cancelled

12. Donor Profile

Edit Profile, Availability, Donation History, Notifications

13. Receiver Module

Create Blood Request

Request Tracking

Matching Donors

Receiver Dashboard

Receiver Profile

14. Notification Module

Push Notifications, SMS, Email

15. Trust & Safety

Report User

16. Blood Group Verification

Medical Report Upload

Admin Verification

17. Availability Rules

Available / Unavailable Matching Logic

18. Location & Matching Engine

Radius:

5KM, 10KM, 20KM, 50KM

19. Admin Web Portal

Dashboard

User Management

Verification Management

Request Management

Donation Management

Notification Management

Reports & Analytics

20. Database Tables

Users

Roles

AadhaarVerification

EmailVerification

BloodVerification

Availability

BloodRequests

DonorMatches

Donations

DonationHistory

Notifications

Reports

AuditLogs

Settings

Hospitals

21. APIs

Authentication APIs

User APIs

Donor APIs

Receiver APIs

Notification APIs

Admin APIs

22. Security

JWT Authentication

RBAC

Audit Logs

Encryption

Rate Limiting

Session Timeout

Database Schema

DATABASE SCHEMA

Users

- UserId (PK)
- FullName
- Mobile
- Email
- AadhaarNumber
- BloodGroup
- RoleId
- City
- Area
- Status
- CreatedAt

Roles

- RoleId (PK)
- RoleName

BloodRequests

- RequestId (PK)
- ReceiverId (FK)
- PatientName
- HospitalName
- BloodGroup
- UnitsRequired
- Urgency
- Status
- CreatedAt

Donations

- DonationId (PK)
- RequestId (FK)
- DonorId (FK)
- Status
- DonationDate

DonationHistory

- HistoryId (PK)
- DonationId (FK)
- HospitalName
- PatientName
- Status

Notifications

- NotificationId (PK)
- UserId (FK)
- Title
- Message
- Status
- CreatedAt

Availability

- AvailabilityId (PK)
- UserId (FK)
- IsAvailable
- UpdatedAt

BloodVerification

- VerificationId (PK)
- UserId (FK)
- ReportFile
- Status

AuditLogs

- LogId (PK)
- UserId (FK)
- Action
- Timestamp